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(54) **DISPLAY ASSEMBLIES WITH VENTS**

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See application file for complete search history.

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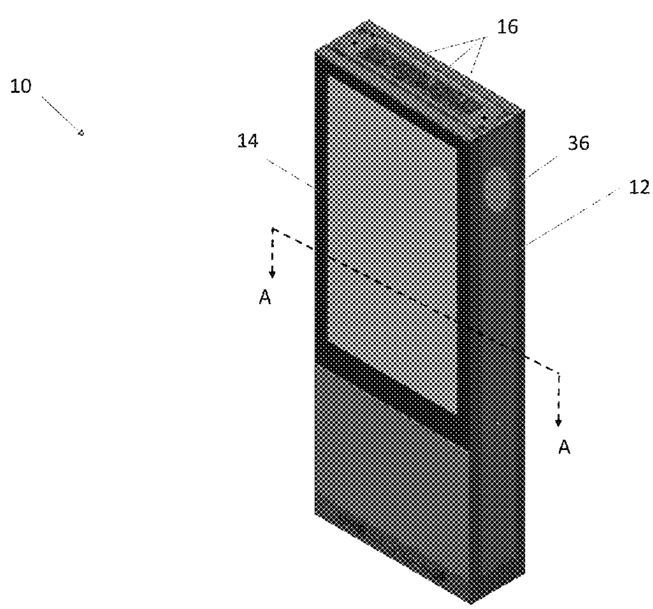
(57) **ABSTRACT**

An electronic display assembly with ventilation, and systems and methods related to the same are provided. The electronic display assembly includes an electronic display layer, a closed loop airflow pathway, and one or more vents fluidly interposed between the closed loop airflow pathway and an ambient environment. A controller may control the vents selectively and individually between an open state and a closed state.

20 Claims, 8 Drawing Sheets



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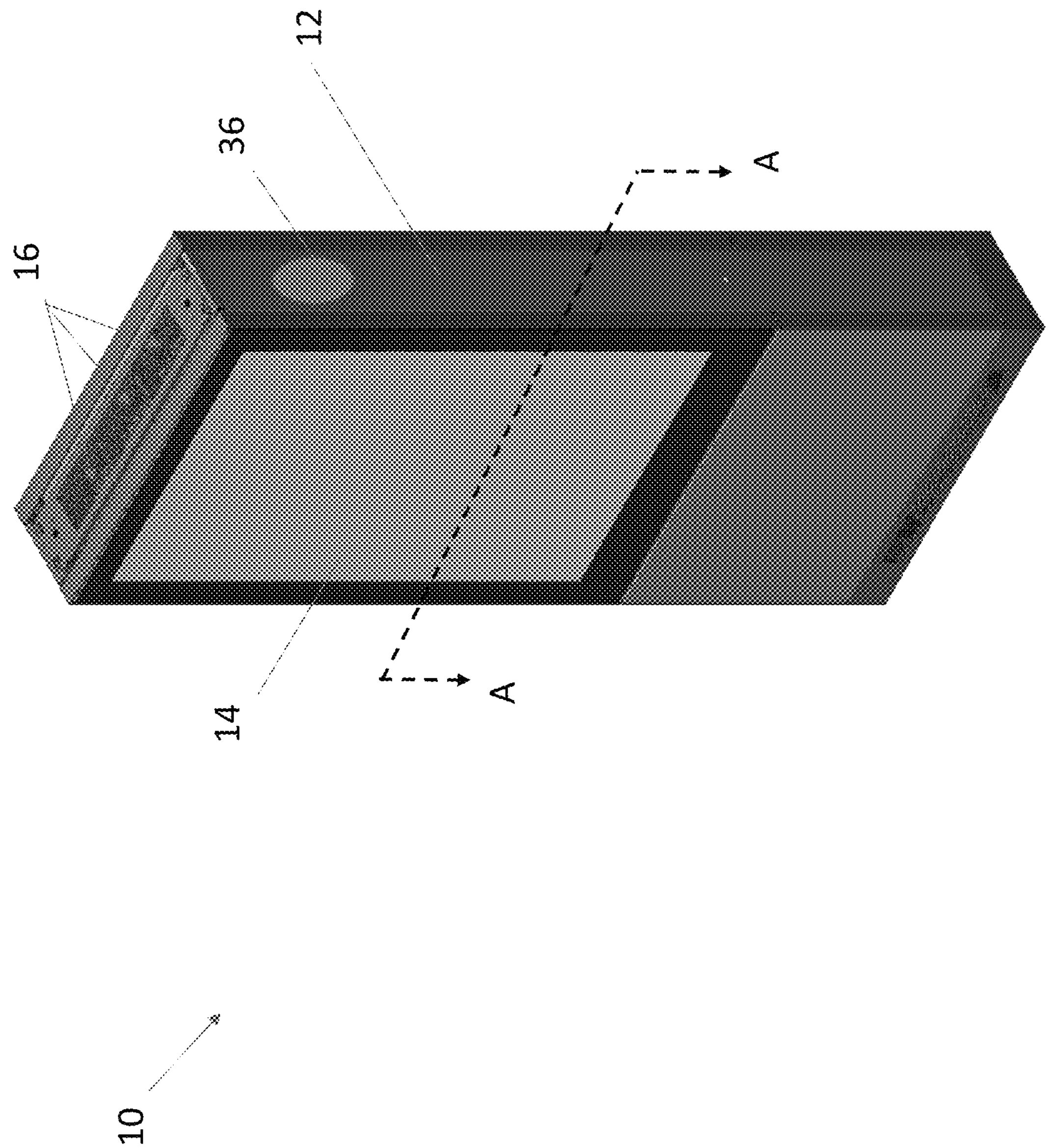
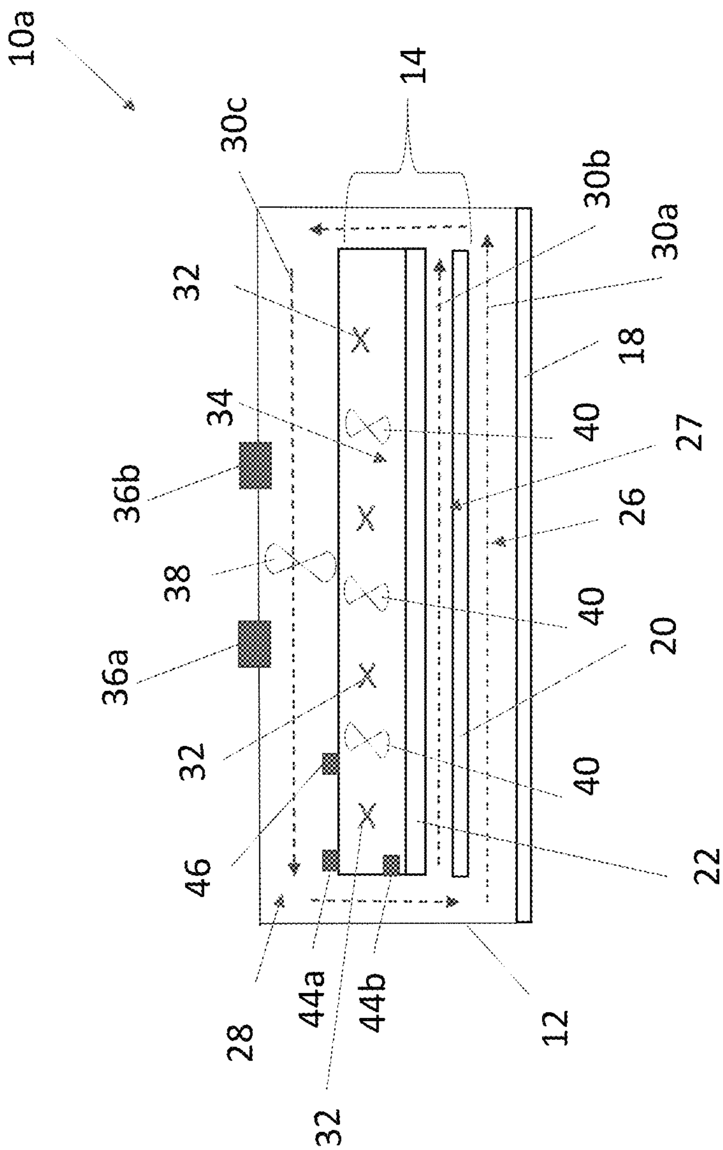


Figure 1

Figure 2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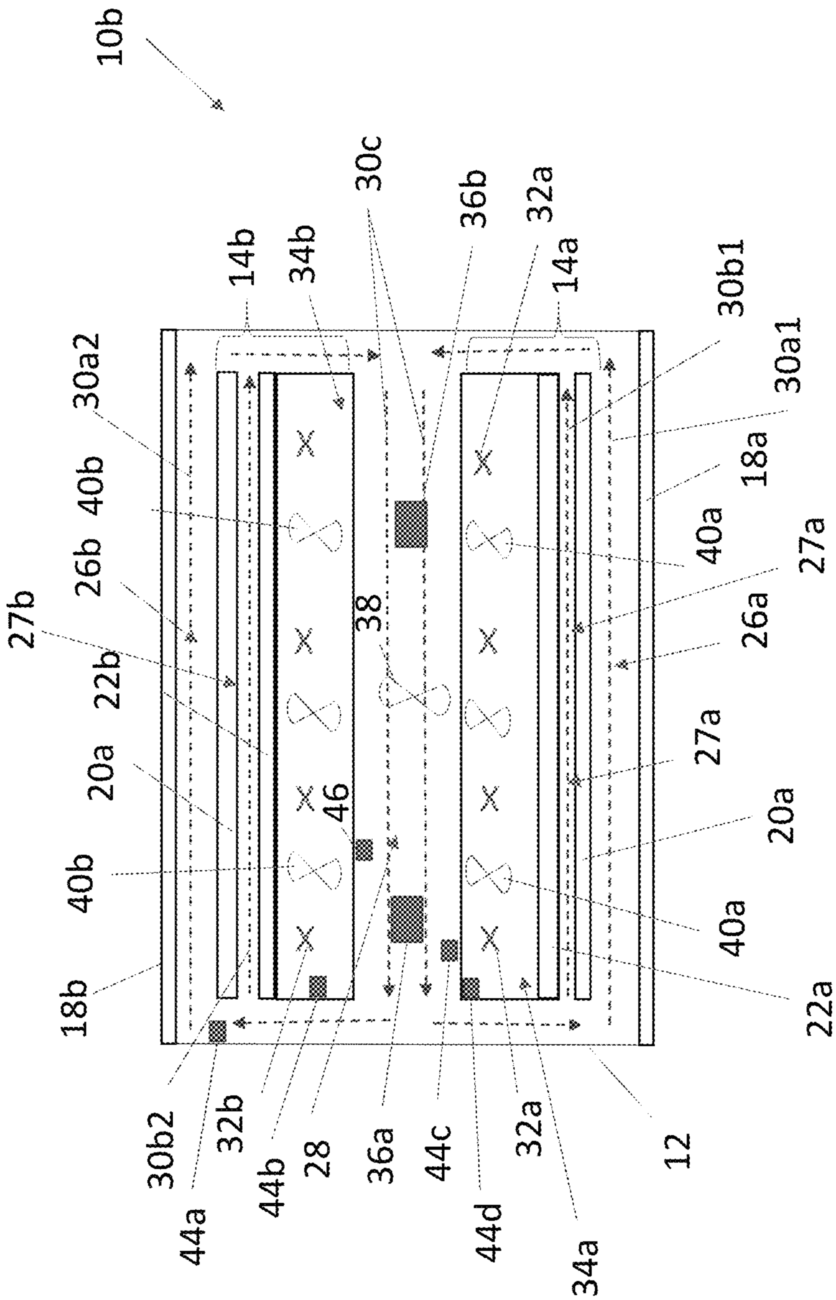
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32b
44b
28
36a
44c
44d
32a
12
44a
34a

Figure 3

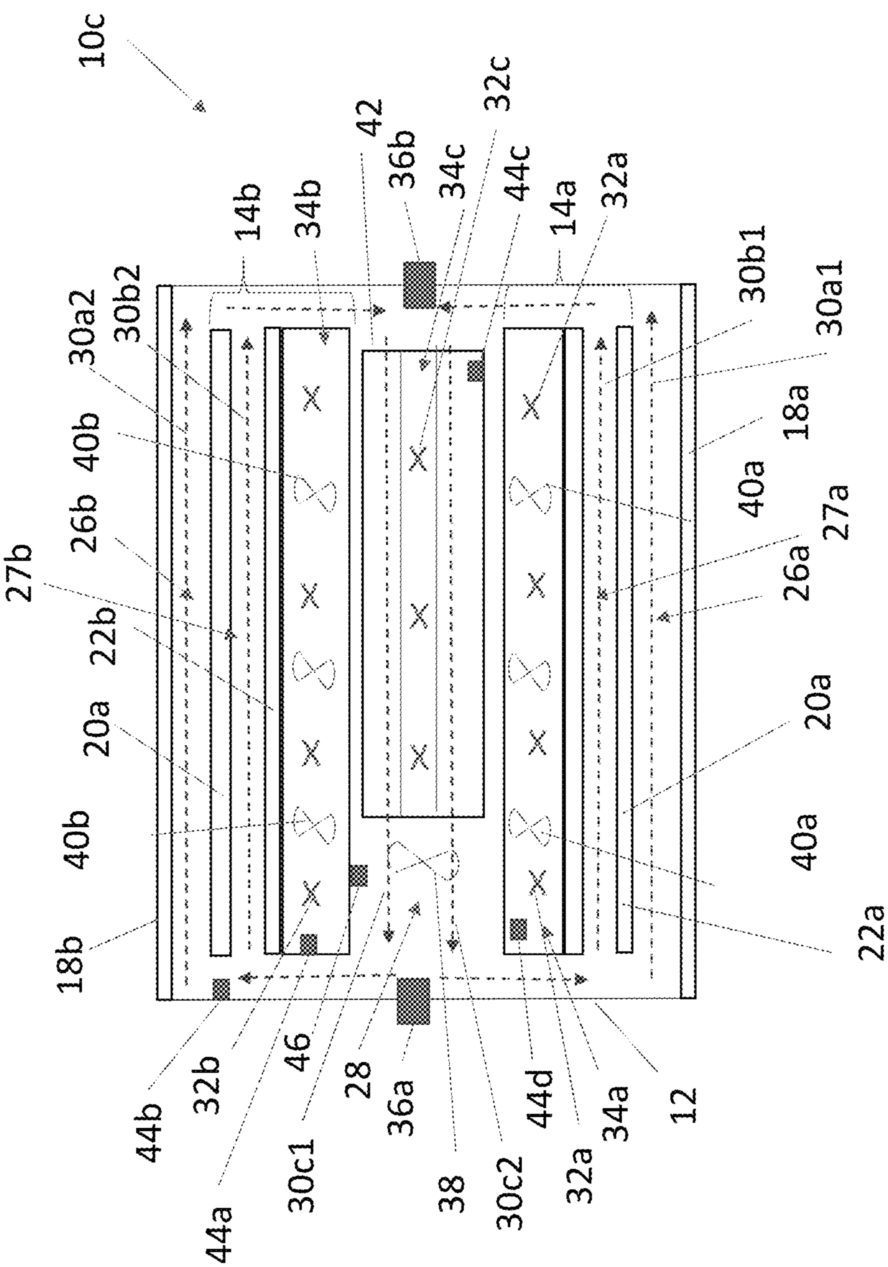

10c
42
32c
36b
34c
44c
14a
32a
14b
34b
30b1
30a1
30a2
30b2
18a
40b
40a
27a
26b
26a
27b
22b
20a
20a
40b
40a
18b
22a
44b
32b
46
28
36a
44d
34a
12
44a
30c1
38
30c2
32a

Figure 4

Figure 5

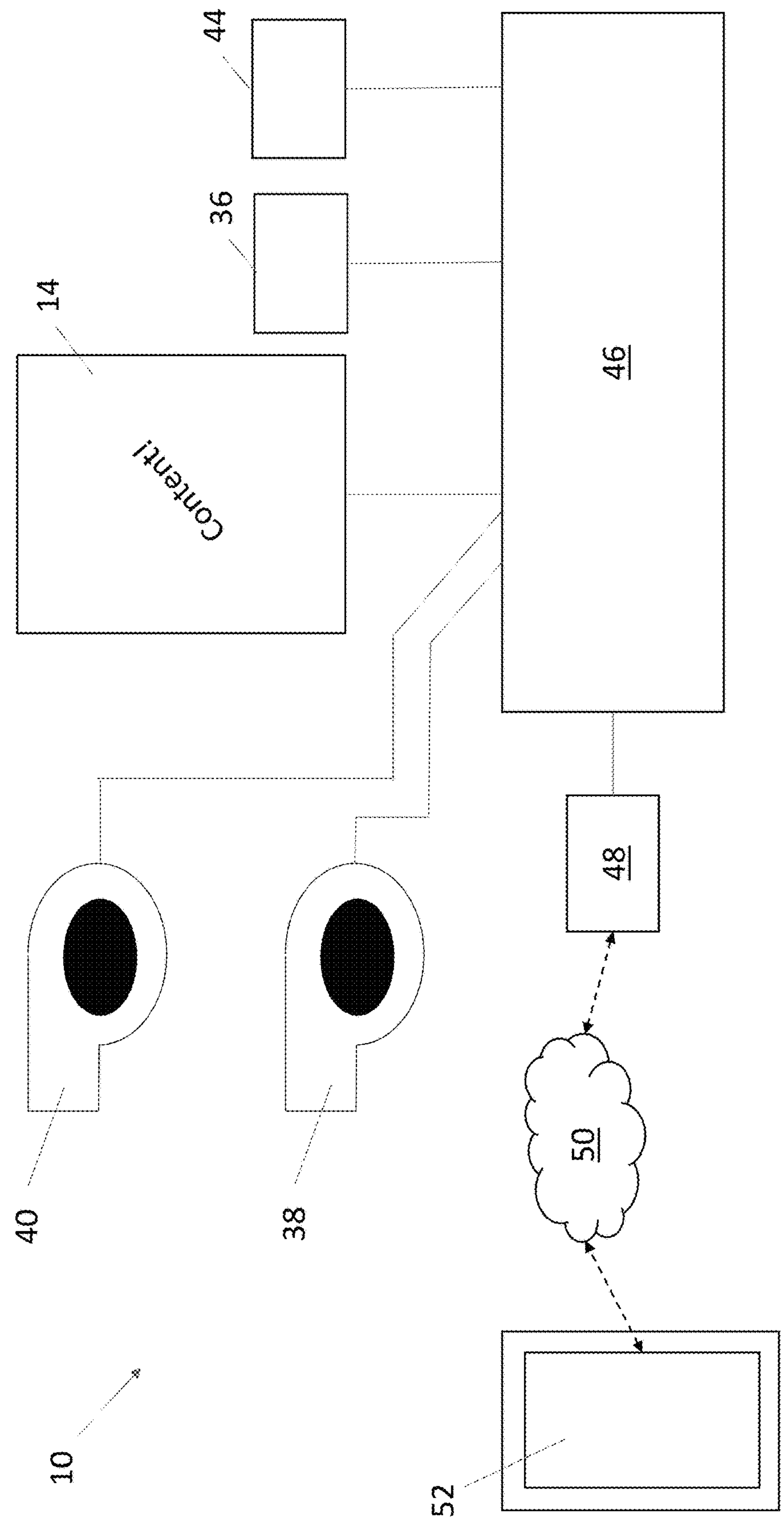


Figure 6

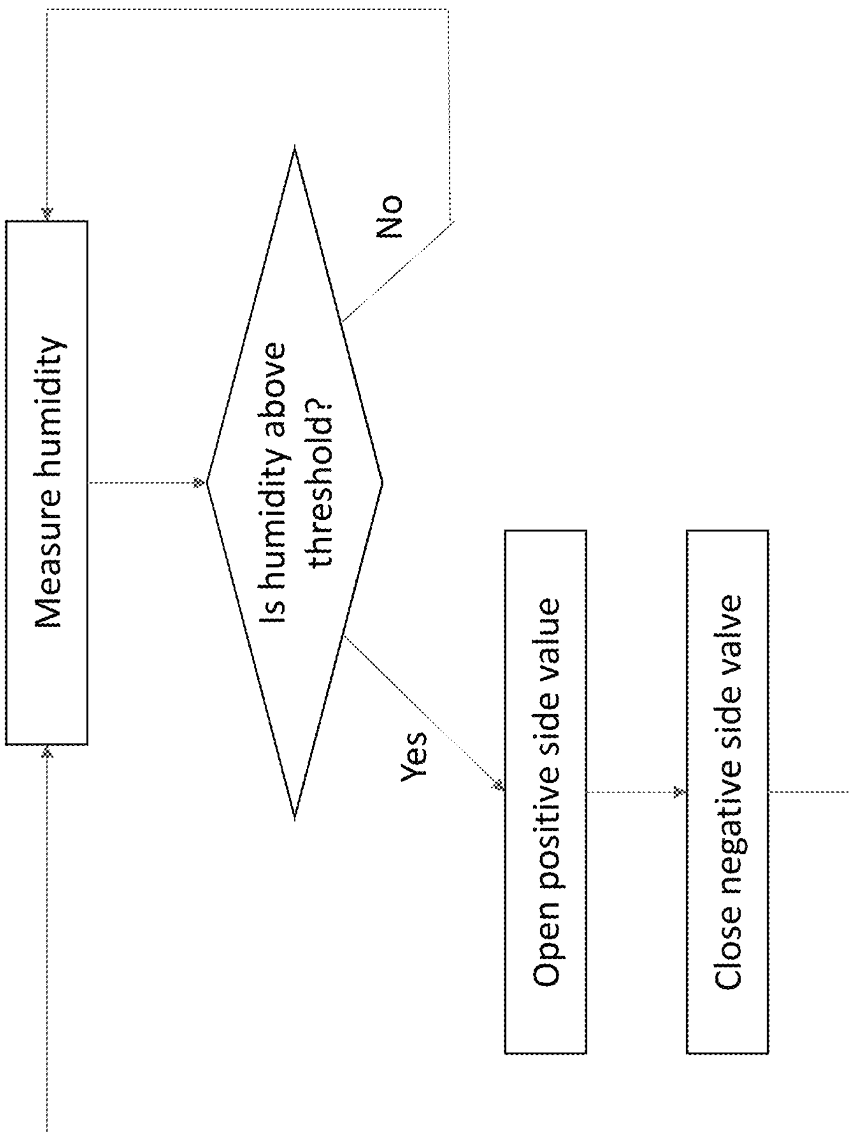

Measure humidity
Is humidity above threshold?
No
Yes
Open positive side value
Close negative side valve

Figure 7

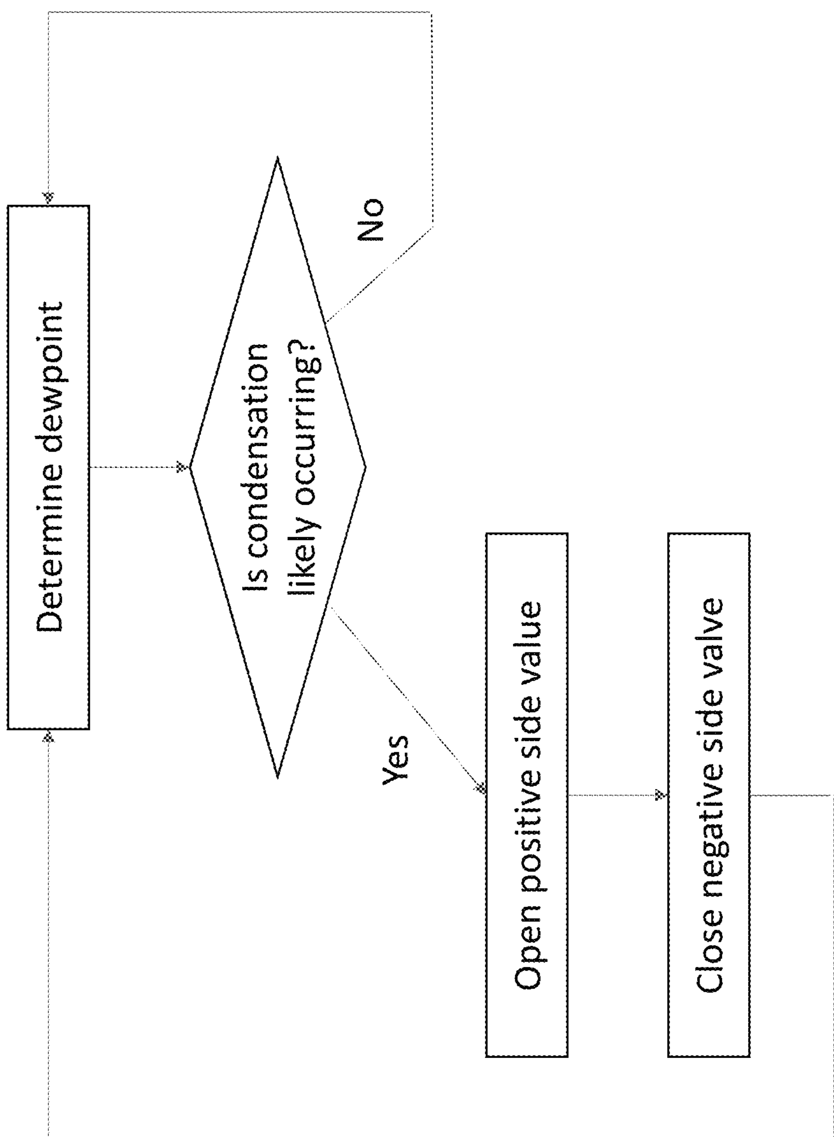
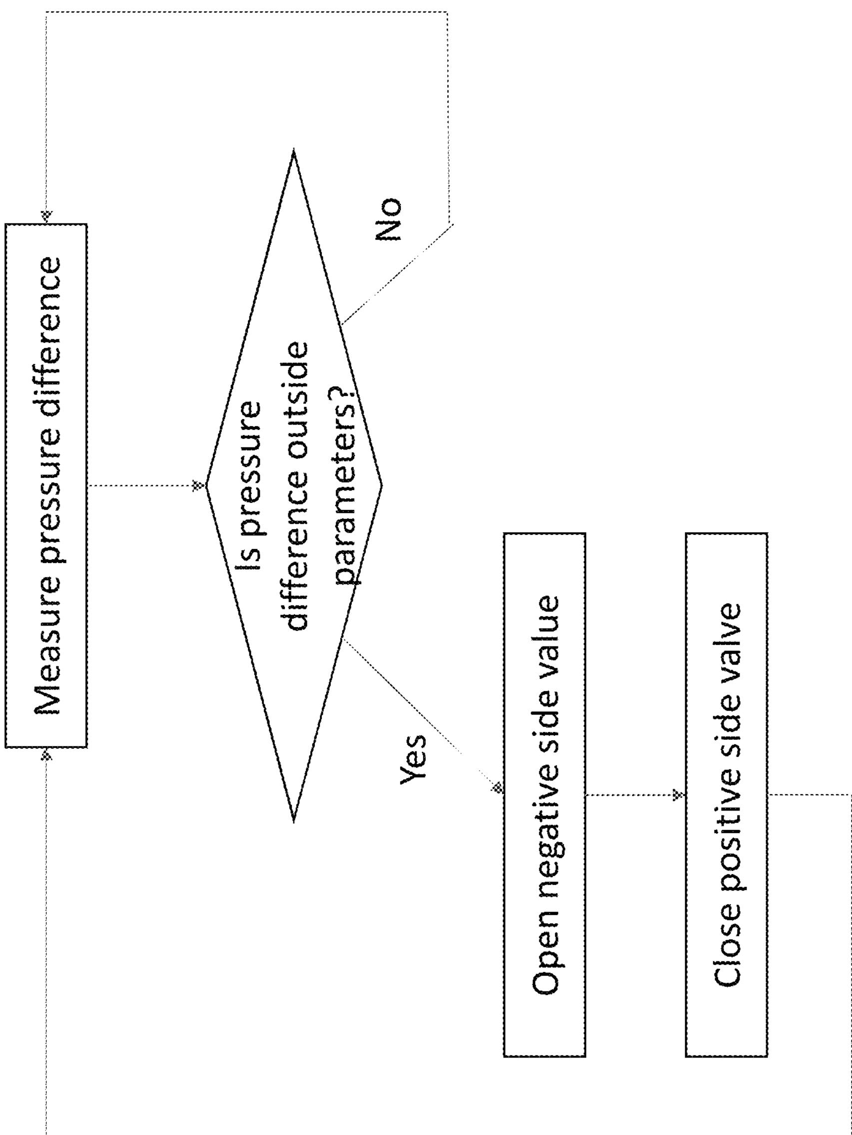

Determine dewpoint
Is condensation likely occurring?
No
Yes
Open positive side value
Close negative side valve

Figure 8


Measure pressure difference
Is pressure difference outside parameters?
No
Yes
Open negative side value
Close positive side valve

DISPLAY ASSEMBLIES WITH VENTS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. provisional patent application No. 63/226,314 filed Jul. 28, 2021, the disclosures of which are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

Exemplary embodiments relate generally to display assemblies with vents, such as for humidity and/or pressure control.

BACKGROUND AND SUMMARY OF THE INVENTION

The use of electronic displays, such as for advertising, in the out-of-home market has increased in popularity over recent years. Being located outdoors, such electronic displays are frequently exposed to harsh conditions, including, but not limited to, solar loading, extreme temperatures, precipitation, moisture, contaminants, vandalism, wildlife, and the like. To protect the electronic displays, and associated sensitive components, from such harsh conditions, it is known to place the electronic displays in ruggedized housings. Such housings may fully or partially seal the electronic displays and other associated sensitive components.

It is known to thermally manage such electronic display assemblies using ambient air and/or circulating gas. Such ambient air may pass through one or more open loop airflow pathways within the assembly, and may thermally interact with circulating gas in one or more closed loop airflow pathways within the assembly where such closed loop pathways are used to remove heat generated by the electronic display assemblies, such as at the backlight.

Such electronic display assemblies, particularly those which incorporate fully or partially sealed areas, sometimes experience moisture, such as in the form of humidity, within the interiors of such units, even within the fully or partially sealed area(s), as such area(s) are generally not 100% sealed in a gas impermeable manner. For example, such areas may be partially or fully sealed against contaminants above a given size (e.g., dust or other particulate) and/or liquids, or in accordance with various standards such as IP65, but may still permit at least some gaseous transfer. Such gaseous transfer may bring moisture with it, such as in the form of humidity, potentially resulting in the formation of condensation on interior surfaces of the assemblies. This may result in undesirable fogging or obfuscation of the displayed images, or water damage to sensitive electronic equipment to name some examples.

Additionally, it may be desirable to control air pressure within such electronic display assemblies for various reasons. This may include controlling bowing of electronic displays, wear and tear on seals, or forces experienced on other various sensitive items within the electronic display assemblies. For example, as electronic displays are increasing in size, the large display layers may be subject to bowing or other distortion based on air pressures encountered. In the case of liquid crystal displays, negative pressure on either side of the layer of liquid crystals may result in cell breach and/or color distortion, such as brown mura. As another example, repeated positive and negative net pressure on gaskets or other seals may result in damage overtime. Therefore, what is needed is a system and method for providing selective venting of an electronic display assembly, such as for humidity and/or pressure control.

Systems and methods are disclosed herein which provide selective venting within an electronic display assembly, such as for humidity and/or pressure control. The electronic display assemblies may include one or more vents. Such vents may be installed within the assemblies in a manner which connects the vents between certain fully or partially sealed areas (hereinafter also the "closed loop areas") and an ambient environment. The vents may be configured to permit circulating gas within the closed loop areas with which the respective vent is connected to only enter, or only exit, the closed loop areas. This may be by design, or may be selectively controlled to operate in such a manner. In particular, as electronic display assemblies become better sealed, the need for ventilation may be required to prevent large pressure fluctuations or levels which may otherwise cause mechanical damage.

The vents may be positioned relative to one or more fan assemblies or other airflow features such that they are located within one or more areas normally experiencing net positive, or relatively high, air pressure of circulating gas within such closed loop areas, or within one or more areas normally experiencing net negative, or relatively low, air pressure of the circulating gas. In exemplary embodiments, the vents are placed within such high/positive air pressure areas and may be configured to vent circulating gas within the closed loop areas to the ambient environment, such as when experiencing certain net positive pressures and/or pressures above a certain threshold. In this manner, humidity or other moisture within the assemblies may be vented with the expelled circulating gas to reduce or eliminate such humidity or other moisture within the assemblies. Alternatively, or additionally, the vents placed within such low/negative air pressure areas may be configured to vent ambient air **32** from the ambient environment into the closed loop areas, such as when experiencing certain net negative pressures and/or pressures below a certain threshold. In this manner, air pressure within the closed loop areas may be raised to, or maintained at, net positive pressure and/or above a certain threshold or relative to other portions of the unit, such as to keep the electronic display layer in compression, to keep humidity and/or other particulate out of the closed loop areas, combinations thereof, or the like. Particularly in the case of liquid crystal displays, compression on the electronic display layer may help to prevent cell breach, brown mura, or other visual distortions. As another example, without limitation, such positive or higher pressures may assist with improving optical performance, such as by preventing distortion by bowing, for example without limitation, of any type or kind of electronic display layer. As yet another example, such positive or higher pressures may assist with driving humidity and other contaminates out of the units.

Opening and closing of the vents may be provided automatically by design of the vents, such as but not limited to, by way of one or more mechanical features of the vents which are designed to operated when experiencing particular absolute or relative pressures. Alternatively, or additionally, opening and closing of the vents may be controlled electronically by one or more controllers. Such opening and closing, by design or electronic operation, may be performed in response to various operating conditions, such as but not limited to, measurements from one or more humidity sensors, temperature sensors, pressure sensors, fan speed sensors, relative pressures experienced at the vents, absolute pressures experienced at the vents, combinations thereof, or the like. In this manner, the vents may be designed and/or controlled to reduce or eliminate condensation, control humidity, and/or maintain positive air pressure within the closed loop areas.

Alternatively, or additionally, the vents may be configured to naturally permit a certain amount of flow from a relatively high-pressure area to a relatively low-pressure area. For example, without limitation, the vents may comprise one or more openings and/or fluid passageways configured to permit various volumetric flow rates at various relative pressures.

Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

FIG. **1** is a perspective view of an exemplary electronic display assembly in accordance with the present invention also illustrating section line A-A;

FIG. **2** is a simplified sectional view of an exemplary embodiment of the electronic display assembly of FIG. **1** taken along section line A-A;

FIG. **3** is a simplified sectional view of another exemplary embodiment of the electronic display assembly of FIG. **1** taken along section line A-A;

FIG. **4** is a simplified sectional view of another exemplary embodiment of the electronic display assembly of FIG. **1** taken along section line A-A;

FIG. **5** is a simplified electrical schematic for use with the electronic display assembly of FIG. **1**;

FIG. **6** is a flow chart with exemplary logic for operating the electronic display assembly of FIG. **5**;

FIG. **7** is a flow chart with other exemplary logic for operating the electronic display assembly of FIG. **5**; and

FIG. **8** is a flow chart with other exemplary logic for operating the electronic display assembly of FIG. **5**.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configuration and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

FIG. **1** is a perspective view of an exemplary electronic display assembly (hereinafter also a "unit") **10** in accordance with the present invention. The unit **10** may include a structural framework **12**. The structural framework **12** may be configured for mounting to a ground surface, such as a sidewalk or street, mounting to a wall or other surface, incorporation into street furniture (e.g., phone booths, bus shelters, benches, railings, combinations thereof, or the like), combinations thereof, or the like. The structural framework **12** may comprise one or more members, panels, cladding, combinations thereof, or the like and may be configured to form a complete or partial housing. The structural framework **12** may form a full or partial housing for some or all components of the assembly **10**.

The units **10** may comprise one or more electronic display subassemblies **14**. Some or all of the electronic display subassemblies **14** may be attached to the structural framework **12** in a moveable manner, though such is not required. For example, the electronic display subassemblies **14** may be attached to the structural framework **12** in a hinged manner to permit selective movement between a closed position whereby certain parts of the units **10** are fully or partially sealed, and an open position whereby certain parts of the interior of the unit **10** are exposed for access. Full or partial sealing may be provided in accordance with one or more standards, such as but not limited to, ingress protection (IP) ratings provided by the International Electrotechnical Commission (e.g., available at https://www.iec.ch/ip-ratings), such as but not limited to, IP 63, 64, 65, 66, 67, and/or 68 by way of non-limiting example.

One or more intakes and exhausts **16** may be provided at the units **10** for ingesting and exhausting ambient air **32**.

FIG. **2** through FIG. **4** illustrate various exemplary embodiments of airflow pathways within the units **10**. FIG. **2** illustrates an exemplary unit **10*a*** with a single electronic display subassembly **14**. FIGS. **3** and **4** illustrate exemplary units **10*b***, **10*c*** with two electronic display subassemblies **14*a***, **14*b*** placed in a back-to-back arrangement. Any number of electronic display subassemblies **14** may be utilized in any arrangement with the structural framework **12**. Similar or the same components used in conjunction with units **10** having multiple electronic display subassemblies **14** may use the same numbering with the addition of an "a", "b", "c", and/or "1", "2", etc. (e.g., **14** to **14*a***, **14*b***, **30*a*** to **30*a*1**, **30*a*2**).

Each electronic display subassembly **14** may comprise an electronic display layer **20**. In exemplary embodiments, the electronic display layer **20** of each of the electronic display subassemblies **14** may comprise an LCD type display with a layer of liquid crystals, and each of the illumination devices **22** may comprise a direct backlight or edge lighting. However, other types of electronic display technologies may be utilized for such electronic display subassemblies **14** including, but not limited to, plasma displays, LED displays, OLED displays, rear projection, cathode ray tube, combinations thereof, or the like.

Each electronic display subassembly **14** may comprise an illumination device **22**. In exemplary embodiments, the illumination device **22** may comprise a number of lighting elements, such as LEDs, provided at a substrate. In exemplary embodiments, the illumination device **22** may be provided rearward of the electronic display layer **20** to serve as a direct backlight. In other exemplary embodiments, the illumination device **22** may comprise one or more diffusive and/or transmissive layers and the substrate and/or lighting elements may be positioned about the edge of the electronic display layer **20** to provide edge lighting to the same. In certain exemplary embodiments, such as where the electronic display layer **20** is an LED, OLED, or other type of self-illuminating display, the illumination device **22** may not be required.

The electronic display layer **20** and/or illumination device **22** may be positioned rearward of a cover **18**. The cover **18** may comprise one or more layers of a transparent or translucent material. In exemplary embodiments, each cover **18** may comprise two layers bonded with an optically clear adhesive. One or more polarizers, anti-reflective materials, combinations thereof, or the like may be disposed on some or all of the cover **18**. The cover **18** may form part of the electronic display subassembly **14** or may be separate therefrom. The cover **18** and the structural framework **12** may together substantially enclose the units **10**, such as with intakes/exhausts **16** exempted. The cover **18** may be configured to move with the electronic display subassembly **14**, may be configured for independent movement, and/or may be fixed to the structural framework **12**.

A single or multiple such electronic display subassemblies **14** may be provided at a single unit **10**, such as but not limited to, in a back-to-back arrangement. The electronic display subassemblies **14** may be of the same or different type and may comprise the same or different components. The electronic display subassemblies **14** may be provided in any arrangement such as portrait or landscape.

The intakes and/or exhausts **16** may be fluidly connected to one or more open loop airflow pathways **34** within the units **10**. A respective one of the open loop airflow pathways **34***a*, **34***b* may extend through a respective one of the electronic display subassemblies **14***a*, **14***b* in exemplary embodiments such that an open loop airflow pathway **34** is provided for each one of the electronic display subassemblies **14**, which may be entirely separate or separated for a distance and rejoined. For example, without limitation, the open loop airflow pathways **34** may extend behind and along at least a portion of the illumination devices **22** for the electronic display layers **20** and/or behind and along at least a portion of the electronic display layers **20** itself, such as in the case of LED, OLED, or other self-illuminating display. The open loop airflow pathways **34** may comprise one or more corrugated layers in exemplary embodiments. However, any type, arrangement, and/or number of airflow pathway(s) may be utilized. One or more filters may be provided at the intakes and/or exhausts **16** and/or along the one or more open loop airflow pathways **34** within the units **10**, though such is not necessarily required.

One or more closed loop airflow pathways may be provided within the units **10**. In exemplary embodiments, such closed loop airflow pathways may comprise at least a front chamber **26**, which may extend between the cover **18** and the electronic display layer **20**, and a rear chamber **28**, which may extend behind the electronic display subassembly **14**, or at least the electronic display layer **20**, but within the structural framework **12**. Where multiple electronic display subassemblies **14***a*, **14***b* are utilized, the rear chamber **28** may be common to each of the electronic display subassemblies **14***a*, **14***b*. A heat exchanger **42** may be located within the rear chamber **28**, though such is not required. The heat exchanger **42** may comprise a multilayer heat exchanger configured to accommodate a common flow **30***c* of the circulating gas through at least some of the layers as well as one or more flows **32***c* of ambient air through at least some other ones of the layers **34***c*. However, any type, arrangement, and/or number of airflow pathway(s) may be utilized. One or more filters may be provided at or along the one or more closed loop airflow pathways within the units **10**, though such is not necessarily required. Filters, barriers, walls, gaskets, combinations thereof, or the like may provide separation between open and closed loop airflow pathways.

An illumination device chamber **27** may extend between each of the electronic display layers **20***a*, **20***b* and the respective illumination devices **22***a*, **22***b*. A flow of circulating gas **30***c* within the rear chamber **28** may be separated such that a first portion **30***a* flows through the front chamber **26** and a second portion **30***b* flows through the illumination device chamber **27**. The flows **30***a*, **30***b* may be recombined, such as in the rear chamber **28**.

One or more open loop fans **40** may be provided. The same of different open loop fans **40** may be associated with each of the open loop airflow pathways **34**. The open loop fans **40** may be configured to ingest ambient air **32** into the units **10**, exhaust ambient air **32** from the assembly **10**, and/or move ingested ambient air **32** through the one or more open loop airflow pathways **34** when activated. One or more closed loop fans **38** may be provided. The same or different closed loop fans **38** may be associated with each of the closed loop airflow pathways. The closed loop fans **20** may be configured to move circulating gas through said one or more closed loop airflow pathways when activated. The fans **38**, **40** may comprise axial fans, centrifugal fans, combinations thereof, or the like. Any number or type of fans **38**, **40** may be used at any location in the units **10**, and may be provided in banks or sets. The open loop airflow pathways **34** may be separate from the closed loop airflow pathways, though a complete (e.g., gas impermeable) separation is not necessarily required.

Examples of such airflow configurations and/or operations may include, for example without limitation, those shown and/or described in one or more of U.S. Pat. No. 8,854,595 issued Oct. 7, 2014, U.S. Pat. No. 8,767,165 issued Jul. 1, 2014, U.S. Pat. No. 8,654,302 issued Feb. 18, 2014, U.S. Pat. No. 8,351,014 issued Jan. 8, 2013, U.S. Pat. No. 10,660,245 issued May 19, 2020, U.S. Pat. No. 10,194,564 issued Jan. 29, 2019, and/or U.S. Pat. No. 10,398,066 issued Aug. 27, 2019, the disclosures of each of which are hereby incorporated by reference in their entireties. The structure and/or mechanical operation of the units **10**, and the various components thereof, and/or airflow configurations may include those shown and/or described in U.S. Pat. No. 10,485,113 issued Nov. 19, 2019 (the "'133 Patent"), the disclosures of which are hereby incorporated by reference in their entirety. Movement and/or structure for facilitating movement of the electronic display subassemblies **14** may be as shown and/or described in at least the '133 Patent.

The unit **10** may comprise one or more controllers **46**. The controller(s) **46** may comprise one or more programmable logic devices. The unit **10** may comprise one or more sensors **44**. The sensors **44** may comprise, for example without limitation, temperature sensors, fan speed sensors, airflow sensors, humidity sensors, relatively humidity sensors, air pressure sensors, differential pressure sensors, location sensors, moisture sensors, combinations thereof, or the like. Any type, kind, or number of sensors **44** may be utilized at any number of locations within the units **10**. The sensor(s) **44** may be in electronic communication with the controller(s) **46**.

Each of the units **10** may comprise one or more vents **36**. The vents **36** may be configured for one way operation, such as but not limited to, to only permit circulating gas to vent from a fluidly connected area of the closed loop airflow pathway(s) to the ambient environment, or only permit ambient air **32** from the ambient environment to vent into the fluidly connected area of the closed loop airflow pathway(s). However, two-way vents (ingestion and exhaustion) may alternatively, or additionally, be utilized. Opening and closing of the vents **36** may be configured to occur under certain conditions, such as but not limited to, at certain air pressures, by design and/or may be electronically controlled. For example, without limitation, the vents **36** may be configured to automatically open or close to permit ingestion of ambient air into the closed loop areas and/or exhaustion of circulating gas from the closed loop areas mechanically when experiencing certain conditions, such as but not limited to, certain relative pressure differences between ambient air and circulating gas and/or certain absolute pressures of ambient air and/or circulating gas, based on physical design and/or structure of the vent **36**. Alternatively, or additionally, the vents **36** may be configured to be operated mechanically based on electronic command, such as but not limited to, by one or more motors.

In exemplary embodiments, each of the vents **36** may be connected to one or more areas of one or more closed loop airflow pathway(s) within the unit **10**, and the ambient environment directly or to one of more of the open loop airflow pathways **34**. Such connections may be direct, such as where the vent **36** extends between such environments, or indirect, such as by way of tubing or other fluid passageways. The vents **36** may connect to the ambient environment directly and/or by way of the one or more of the open loop airflow pathways **34**.

Some or all of the vents **36** may be passive vents. The vents **36** may be configured to naturally permit flow of air from relatively high-pressure areas to relatively low-pressure areas. Some or all of the vents **36** may comprise fluid passageways such that the vents **36** are capable of selectively releasing circulating gas and/or ingesting ambient air. The size and/or shape of openings in and/or fluid passageways within the vents **36** may be configured to permit various volumetric flow rates at various relative pressures. The size and/or shape of openings in and/or fluid passageways within the vents **36** may be configured to naturally resist particulate and/or water ingestion. The vents **36** may optionally include one or more filters, membranes, or the like configured to resist or prevent particulate and/or water ingestion and/or permit one way flow of air.

Examples of such vents include, for example without limitation, those available from W. L. Gore & Associates, Inc. of Newark, DE (https://www.gore.com/products/categories/venting) and/or Amphenol Corp., of Wallingford, Ct. (https://www.amphenol.com/), including but not limited to the VEN-PS1NBK-08001 and/or VEN-PS1NBK-08002 products from Amphenol. These are merely exemplary and are not intended to be limiting.

In exemplary embodiments, a first one of the vents **36*a*** may be placed on a first side of one of the closed loop fans **38** which may be associated with relatively high and/or net positive pressures when the fan **38** is operated. A second one of the vents **36*b*** may be placed on a second side of one of the same or different ones of the closed loop fans **38**, such as associated with relatively low and/or net negative pressures when the fan **38** is operated. In other exemplary embodiments, a single or multiple vents **36** may be placed exclusively on one side of the closed loop fans **38**. Any number of vents **36** in any number of locations may be utilized. The vents **36** may be configured to ingest ambient air into, and/or exhaust circulating gas from, the associated closed loop areas. In exemplary embodiments, without limitation, one or more passive type vents **63** is placed in close proximity to a negative pressure side of the closed loop fans **38**. In other exemplary embodiments, without limitation, one or more passive type vents **36** is placed in close proximity to a positive pressure side of the closed loop fans **38**.

In exemplary embodiments, one or more of the vents **36** may be positioned on a side of the closed loop fans **38** associated with negative, or relatively low pressure primarily for ingesting ambient air into the associated closed loop area. In this manner, pressures within the closed loop airflow pathways, including the portions of the flow **30*a*** and **30*b*** within the front chamber **26** and the illumination devices chamber **27** may be raised to, and/or maintained at, positive pressures to place or keep the electronic display layer **20** in compression. This may prevent cell breach and/or color distortion.

Alternatively, or additionally, one or more of the vents **36** may be positioned on a side of the closed loop fans **38** associated with positive, or relatively high pressure primarily for exhausting circulating gas in the associated closed loop area to the ambient environment. In this manner, circulating gas may be periodically driven out of the closed loop areas. This may reduce or eliminate moisture and other contaminants. This may also maintain the electronic display layer **20** in tension, such as to keep it taught.

In exemplary embodiments, placement of, or at least selective opening of, one or more vents **36** on the side of the closed loop fans **38** associated with relatively high and/or net positive pressures when the fan **38** is operated may result in increasing the average pressure of the circulating gas in the associated closed loop area as compared to without such one or more vents **36** and/or keeping such one or more vents **36** closed. In exemplary embodiments, placement of, or at least selective opening of, one or more vents **36** on the side of the closed loop fans **38** associated with negative, or relatively low pressure when the fan **38** is operated may result in decreasing the average pressure of the circulating gas in the associated closed loop area as compared to without such one or more vents **36** and/or keeping such one or more vents **36** closed. In this manner, placement and/or operation of the vent(s) **36** may be controlled to control or influence pressures of circulating gas in the closed loop airflow pathway.

Alternatively, or additionally, the vent(s) **36** may be fluidly connected to such areas (e.g., those on the positive of negative side of the closed loop fans **38**). Such fluid connection may be made by way of tubes, for example without limitation. This may permit the vent(s) **36** to be placed in more convenient locations and fluidly connected to the desired region, such as to accomplish the same, or substantially the same, effect while permitting greater design variation and/or where desired placement is impractical or impossible.

FIG. **5** is a simplified electrical schematic for use with the electronic display assembly **10**. The controller **46** may be in electronic communication with some or all components of the assembly **10**. The controller **46** may be in electronic communication with each of the electronic display subassemblies **14**. The controller **46** may be in electronic communication with each of the closed loop fans **38**. The controller **46** may be in electronic communication with each of the open loop fans **40**. The controller **46** may be in electronic communication with each of the sensors **44**. The controller **46** may be in electronic communication with each of the vents **36**. The controller **46** may be configured to receive data from such components and/or send commands to such components by wired or wireless connection. For example, without limitation, the controller **46** may be configured to open or close the vents **36**, control speeds of the fans **38**, **40**, content displayed at the electronic display subassemblies **14**, take measurements from the sensors **44**, combinations thereof, or the like. Such electronic communication is not required between the controller **46** and the aforementioned components. For example, without limitation, the vents **36** may be configured to operate based on experienced conditions (e.g., pressures) and without the need for electronic control.

The controller **46** may be in electronic communication with a network communication device **48**. The network communication device **48** may be configured to receive data from the controller **46** for transmission over one or more networks **50** to one or more electronic devices **52**, such as to permit remote monitoring of the units **10**. The network communication device **48** may be configured to receive commands from the one or more electronic devices **52** for passing to the controller **46**, such as to permit remote control over the units **10**. The networks **50** may comprise cellular networks, wireless networks, wired networks, combinations thereof, or the like. The electronic devices **52** may comprise personal computers, smartphones, tablets, network operation centers, combinations thereof, or the like.

FIG. **6** is a flow chart with exemplary logic for operating the electronic display assembly **10**. Humidity may be measured by way of at least one of the sensors **44**, which may comprise a humidity sensor and/or relative humidity sensor (e.g., psychrometer). The measurement(s) may be reported to, or determined at, the controller **46** from data received from the at least one of the sensors **44**. If the humidity is above a threshold, the controller **46** may be configured to open at least one of the vents **36**. In exemplary embodiments, the controller **46** may be configured to open one or more of the vents **36** on a positive or relatively high-pressure side of one or more of the closed loop fans **38**. The controller **46** may, alternatively or additionally, be configured to close one or more of the vents **36** on a negative or relatively low-pressure side of one or more of the closed loop fans **38**. In other exemplary embodiments, the controller may be configured to open one or more of the vents **36** configured for one-way expulsion of circulating gas and/or close one or more of the vents configured for one-way ingestion of ambient air. Such operations may occur simultaneously or in sequence. This may permit expulsion of moisture from the units **10** with the expelled circulating gas. The threshold may be any amount. In exemplary embodiments, the threshold is set to reflect a point in which condensation is likely to form within the unit **10**, with or without a margin of safety.

FIG. **7** is a flow chart with other exemplary logic for operating the electronic display assembly **10**. Relative humidity may be measured by way of at least one of the sensors **44**. Air temperature may be measured by way of the same or different one(s) of the sensors **44**. These measurements may be reported to, or determined at, the controller **46** from data received from the sensor(s) **44**. Alternatively, or additionally, such measurements may be taken from one or more internet-based sources, such as by way of the network communication device **48**. The controller **46** may be configured to determine a dewpoint for the measured air, which may be air within the units **10** or ambient, and calculate a dewpoint spread relative to the air temperature measurements, which may be ambient air within the units **10** or ambient air outside the units **10**. In exemplary embodiments, dewpoint is calculated for circulating gas within the unit **10** and dewpoint spread is measured relative to ambient air outside of, or within, the units **10**.

Where dewpoint spread reaches one or more thresholds, such as but not limited to within 5° C., the controller **46** may be configured to open one or more of the vents **36** on a positive or relatively high-pressure side of one or more of the closed loop fans **38**. The controller **46** may, alternatively or additionally, be configured to close one or more of the vents **36** on a negative or relatively low-pressure side of one or more of the closed loop fans **38**. Alternatively, or additionally, the controller **46** may be configured to operate such vents **36** in a manner which only permits circulating gas to be expelled from the unit **10** to ambient. Such operations may occur simultaneously or in sequence. In this manner, moisture may be expelled from the units **10** with the expelled circulating gas. The dewpoint spread threshold may be any amount.

FIG. **8** is a flow chart with other exemplary logic for operating the electronic display assembly **10**. Air pressure readings of circulating gas, such as but not limited to, the portion **30***a* of circulating gas within the front chamber **26** and/or the portion **30***b* of the circulating gas within the illumination devices chamber **27** may be measured by way of one or more of the sensors **44**. The measurements may be reported to, or determined at, the controller **46** from data received from the sensors **44**. The controller **46** may be configured to determine if the pressures differential between the first and second portions **30***a*, **30***b* is within certain operating conditions. Such operating conditions may be, for example without limitation, both positive. Alternatively, or additionally, such operating conditions may be, without limitation, higher within the front chamber **26** relative within the illumination device chamber **27**. These operating conditions may be configured to reduce or eliminate bowing of the electronic display layer **20**, and/or maintain compressive forces on the electronic display layer **20**. Where such pressure falls outside of desired conditions, the vents **36** may be selectively operated to control the same. For example, without limitation, one or more of the vents **36** associated with negative or relatively low pressure may be opened and/or configured for ingestion of ambient air so as to raise pressures within the closed loop airflow pathway.

While electronic control of the vents **36** is contemplated in at least certain embodiments, such as but not limited to, those shown and/or described with regard to FIGS. **5**-**8**, the vents **36** may be configured to automatically operate by mechanical design in response to various conditions, such as pressures or relative pressures, experienced and thus such electronic control may not be required. The vents **36** may be configured, for example without limitation, to automatically operate mechanically by engineered deign and without physical control, in response to the conditions shown and/or described herein and/or achieve the results (e.g., humidity reduction, pressure miniatous) shown and/or described herein.

Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

Certain operations described herein may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing device. The electronic devices may comprise personal computers, smartphone, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein.

What is claimed is:

1. An electronic display assembly with ventilation comprising:
  - a structural framework;
  - an electronic display subassembly attached to the structural framework and comprising a cover layer forming a forward surface of said electronic display subassembly and an electronic display layer located behind the cover layer;
  - an enclosure for at least the electronic display layer, said enclosure defined, at least in part, by the structural framework and the cover layer;
  - a closed loop airflow pathway extending within the enclosure; and
  - one or more vents fluidly interposed between the closed loop airflow pathway and an ambient environment, wherein said one or more vents are movable between an open state and a closed state.

2. The electronic display assembly of claim 1 wherein:
  - said one or more vents comprise openings and fluid passageways configured to permit gaseous flows at certain volumetric flow rates at certain pressures to the exclusion of gaseous flows at other volumetric flow rates and other pressures.

3. The electronic display assembly of claim 2 wherein:
  - said one or more vents comprise one or more membranes configured to resist ingestion of particulate and water.

4. The electronic display assembly of claim 2 further comprising:
  - at least one fan located along the closed loop airflow pathway, wherein the said one or more vents are located on a negative pressure side of the at least one fan.

5. The electronic display assembly of claim 1 further comprising:
  - a controller in electronic communication with the one or more vents and configured to selectively and individually operate said one or more vents between the open state and the closed state.

6. The electronic display assembly of claim 5 further comprising:
  - at least one fan located along the closed loop airflow pathway;
  - a first one of the one or more vents located on a first side of said at least one fan associated with a positive pressure; and
  - a second one of the one or more vents located on a second side of said at least one fan associated with a negative pressure.

7. The electronic display assembly of claim 5 further comprising:
  - one or more sensors in electronic communication with said controller, wherein said controller is configured to operate said one or more vents between said open state and said closed state at least in part based on data received from said one or more sensors.

8. The electronic display assembly of claim 7 wherein:
  - said one or more sensors comprise a differential pressure sensor fluidly connected to said closed loop airflow pathway and the ambient environment; and
  - said controller is configured to operate the first one of said vents in said closed state and said second one of said vents in said open state after determining, based on data received from said differential pressure sensor, that a measured pressure difference is outside a target range.

9. The electronic display assembly of claim 7 wherein:
  - said one or more sensors comprise a humidity sensor; and
  - said controller is configured to operate the first one of said vents in said open state and said second one of said vents in said closed state after determining, based on data received from said differential pressure sensor, that a measured humidity is above a predetermined threshold.

10. The electronic display assembly of claim 7 wherein:
  - said one or more sensors comprise a temperature sensor located at the closed loop airflow pathway; and
  - said controller is configured to:
    - measure a temperature of circulating gas within the closed loop airflow pathway by way of the temperature sensor;
    - determine a dewpoint of the ambient environment;
    - determine a dewpoint spread between the temperature and the dewpoint; and
    - where the dewpoint spread is below a predetermined threshold:
      - operate the first one of said vents in said open state; and
      - operate the second one of said vents in said closed state.

11. The electronic display assembly of claim 1 wherein:
  - a first portion of said closed loop airflow pathway extends between said cover and said electronic display layer, a second portion of said closed loop airflow pathway extends rearward of said electronic display layer;
  - said electronic display layer comprises liquid crystals; and
  - said electronic display subassembly comprises a backlight located rearward of the electronic display layer.

12. The electronic display assembly of claim 11 further comprising:
  - a second electronic display subassembly attached to said structural framework, wherein the electronic display subassembly and the second electronic display subassembly are movably attached to opposing sides of said structural framework.

13. The electronic display assembly of claim 1 wherein:
  - each of the one or more vents comprise one-way vents configured to permit exhaust of gas from the electronic display assembly.

14. An electronic display assembly with controllable ventilation, said electronic display assembly comprising:
  - a structural framework;
  - an electronic display subassembly attached to said structural framework and comprising a cover and an electronic display layer;

a closed loop airflow pathway for circulating gas extending within one or more of the structural framework and the electronic display subassembly;

a fan located along the closed loop airflow pathway;

a first controllable vent located on a first side of said fan and fluidly interposed between the closed loop airflow pathway and an ambient environment;

a second controllable vent located on a second side of said fan fluidly interposed between the closed loop airflow pathway and the ambient environment, wherein the first and second controllable vents comprise one-way vents configured to permit exhaust of the circulating gas from the closed loop airflow pathway to the ambient environment;

one or more sensors, at least one of which is located along the closed loop airflow pathway, wherein said one or more sensors are selected from the group consisting of: temperature sensors, humidity sensors, relatively humidity sensors, and pressure sensors; and

a controller in electronic communication with the first and second controllable vents and the one or more sensors, wherein said controller is configured to separately operate said first and second controllable vents between an open state whereby the circulating gas is permitted to exit the closed loop airflow pathway to the ambient environment and a closed state whereby the circulating gas is prevented from exiting the closed loop airflow pathway to the ambient environment based on readings from said one or more sensors.

15. The electronic display assembly of claim **14** further comprising:

an open loop airflow pathway for ambient air extending within one or more of the structural framework and the electronic display subassembly, wherein said first and second controllable values are fluidly connected to the ambient environments by way of the open loop airflow pathway, and wherein at least one of said one or more sensors is located at the open loop airflow pathway.

16. A method for ventilating an electronic display assembly in a controllable fashion, said method comprising:

displaying images at an electronic display layer of said electronic display assembly;

operating a fan located along a closed loop airflow pathway of the electronic display assembly;

receiving data, at a controller, from a sensor located along the closed loop airflow pathway; and

operably controlling, by way of the controller, a first vent located at a first side of said fan and a second vent located at a second side of the fan between an open state and a closed state, wherein each of said first and second vents are fluidly connected to said closed loop airflow pathway and an ambient environment.

17. The method of claim **16**:

wherein said sensor comprises a differential pressure sensor fluidly connected to said closed loop airflow pathway and the ambient environment; and

further comprising:

determining, at the controller by way of data received from the differential pressure sensor, that a measured differential pressure is outside a target range; and

operating, by way of said controller, the first controllable vent in said closed state and said second controllable vent in said open state.

18. The method of claim **16**:

wherein said sensor comprises a humidity sensor; and

further comprising:

determining, at the controller by way of data received from the humidity sensor, that a measured humidity is above a predetermined threshold; and

operating, by way of said controller, the first controllable vent in said open state and said second controllable vent in said closed state.

19. The method of claim **16**:

wherein said sensor comprises a temperature sensor; and

further comprising:

determining, at the controller, a dewpoint for the ambient environment;

determining, at the controller by way of data received from the temperature sensor, a dewpoint spread between the dewpoint and a measured temperature;

determining, at the controller, that the dewpoint spread is less than a predetermined threshold;

operating the first controllable vent in said open state; and

operating the second controllable vent in said closed state.

20. The method of claim **19** wherein:

said dewpoint is determined based on data retrieved from one or more internet-based local weather information sources by way of one or more network communication devices in electronic communication with said controller.

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